

Scalability & Future Plan

Date	26 March 2026
Project Name	Smart Lender

Scalability & Future Plan:

This document outlines how the current project solution can be scaled to handle larger user bases, increased data loads, or extended features in the future. It also captures the team's roadmap for enhancing and evolving the project beyond its current state.

Current System Limitations:

S.No	Limitation	Impact	Priority to Address (High / Medium / Low)
1	Supports only a limited number of concurrent users in the local environment	Reduced performance under heavy load	High
2	No database integration for storing applicant records	Applicant information is not saved permanently	High
3	Basic security features without user authentication and authorization	Limited access control and data protection	Medium

Scalability Plan:

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Supports local users	Deploy on IBM Cloud with load balancing to support multiple concurrent users

2	Data Storage	No persistent database	Integrate MySQL or MongoDB for secure data storage
3	Performance	Local Flask application	Optimize the ML model and deploy on a scalable cloud environment
4	Security	Basic input validation	Implement user authentication, role-based access, HTTPS, and data encryption

Future Roadmap:

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
Phase 1	Add user authentication, database integration, and loan application history	Next 3 Months	Improved security and data management
Phase 2	Deploy the application on IBM Cloud and optimize performance for multiple users	Next 6 Months	Better scalability, reliability, and availability
Phase 3	Integrate explainable AI, email/SMS notifications, and analytics dashboard	Next 12 Months	Enhanced user experience and better decision support